CALCULUS I - AY2024-25

Chapter 5: Techniques of Integration

Outline

1 Integration Techniques

2 Improper Integrals

Summary

In this section we study different techniques to find the anti-derivative of a given continuous function, including

(1) tricks involving trigonometric integrals

(2) trigonometric substitutions

(3) partial fractions.

As already discussed in the previous chapters, two fundamental techniques *"Integration by Substitution"* and *"Integration by Parts"* play an important role in integration.

- Substitution Rule for definite integrals. If $g'(x)$ is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

- Integration by Parts for definite integrals. If f and g are both differentiable, then

$$\int_a^b f(x)g'(x)\,dx = \Big[f(x)g(x)\Big]_a^b - \int_a^b f'(x)g(x)\,dx.$$

(1) Trigonometric Integrals

These are integrals which contain combinations of trigonometric functions.

In order to evaluate trigonometric integrals, we need to know various trigonometric identities - please revise (!).

We study the different trigonometric integrals by listing out all the different cases and showing what we can do in each case.

- **Evaluating $\int \sin^{\mathbf{m}} \mathbf{x} \cos^{\mathbf{n}} \mathbf{x}\, \mathbf{dx}$**

$*$ Case 1. If n is odd, then make the substitution $u = \sin x$.

$*$ Case 2. If m is odd, then make the substitution $u = \cos x$.

$*$ Case 3. If both m, n are even, then reduce powers of $\sin x, \cos x$ by

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin x \cos x = \frac{\sin 2x}{2}.$$

Example

Evaluate

$$(1)\ \int_0^{\frac{\pi}{2}} \sin^2 x \cos^5 x\ dx;\ (2)\ \int_0^{\frac{\pi}{2}} \sin^5 x \cos^2 x\ dx;\ (3)\ \int_0^{\frac{\pi}{2}} \sin^2 x \cos^4 x\ dx.$$

Solution. (1) Case 1; (2) Case (2). *[Answer for both:* $\frac{8}{105}$*]*.

(3) Case 3. Reducing powers, we have

$$\int_0^{\frac{\pi}{2}} \sin^2 x \cos^4 x\ dx = \int_0^{\frac{\pi}{2}} \left(\frac{1-\cos 2x}{2}\right)\left(\frac{1+\cos 2x}{2}\right)^2\ dx$$

$$= \frac{1}{8}\int_0^{\frac{\pi}{2}} \left(1+\cos 2x-\cos^2 2x-\cos^3 2x\right)\ dx$$

Evaluating each integral, we get

$$\int_0^{\frac{\pi}{2}} 1 \, dx = \frac{\pi}{2}, \quad \int_0^{\frac{\pi}{2}} \cos 2x \, dx = 0,$$
$$\int_0^{\frac{\pi}{2}} \cos^2 2x \, dx = \frac{\pi}{4}, \quad \int_0^{\frac{\pi}{2}} \cos^3 2x \, dx = 0.$$

So, we obtain

$$\int_0^{\frac{\pi}{2}} \sin^2 x \cos^4 x \, dx = \frac{1}{8}\left[\frac{\pi}{2} + 0 - \frac{\pi}{4} - 0\right] = \frac{\pi}{32}.$$

- **Evaluating $\int \tan^m x \sec^n x \, dx$**

∗ Case 1. If n is even, then make the substitution $u = \tan x$.

∗ Case 2. If m is odd, and $n \geq 1$, then make the substitution $u = \sec x$.

∗ Case 3. For the other cases, the guidelines are not given. The following formulas are sometimes used

$$\int \tan x \, dx = \ln|\sec x| + C, \qquad \int \sec x \, dx = \ln|\sec x + \tan x| + C.$$

Example

Evaluate

$$(1) \int_0^{\frac{\pi}{4}} \tan^3 x \sec^4 x \, dx; \qquad (2) \int_0^{\frac{\pi}{4}} \tan^3 x \, dx.$$

Solution. (1) Either Case 1 or Case 2 can be applied. *[Answer:* $\frac{5}{12}$*]*.

(3) This does not belong to neither Case 1 (because the power of $\sec x$ is 0) nor Case 2 (because we need at least one copy of $\sec x$ to be present ($n \geq 1$) and for m to be odd).

So what can we do?

Below is one of the possible solutions.

$$\int_0^{\frac{\pi}{4}} \tan^3 x \, dx = \int_0^{\frac{\pi}{4}} \tan x \tan^2 x \, dx = \int_0^{\frac{\pi}{4}} \tan x (\sec^2 x - 1) \, dx$$

$$= \int_0^{\frac{\pi}{4}} \tan x \sec^2 x \, dx - \int_0^{\frac{\pi}{4}} \tan x \, dx$$

$$\underbrace{=}_{(\text{Let } u = \tan x)} \int_0^1 u \, du - \int_0^{\frac{\pi}{4}} \tan x \, dx = \left[\frac{u^2}{2}\right]_0^1 - \left[\ln|\sec x|\right]_0^{\frac{\pi}{4}}$$

$$= \left(\frac{1^2}{2} - \frac{0^2}{2}\right) - \left(\ln\left|\sec\frac{\pi}{4}\right| - \ln|\sec 0|\right)$$

$$= \frac{1}{2} - \left(\ln\sqrt{2} - \ln 1\right) = \frac{1}{2} - \frac{\ln 2}{2}.$$

- **Evaluating products of sine and cosine $\int \sin(\mathbf{mx})\cos(\mathbf{nx})\,\mathbf{dx}$**

To evaluate integrals of the form

$$\int \sin(mx)\cos(nx)\,dx, \quad \int \sin(mx)\sin(nx)\,dx, \quad \int \cos(mx)\cos(nx)\,dx,$$

we use the corresponding identities "products-to-sums":

- $\sin x \cos y = \dfrac{\sin(x+y)+\sin(x-y)}{2}$
- $\cos x \cos y = \dfrac{\cos(x+y)+\cos(x-y)}{2}$
- $\sin x \sin y = \dfrac{\cos(x-y)-\cos(x+y)}{2}$

Example

Evaluate $\int_0^{\frac{\pi}{2}} \sin(3x)\cos(5x)\,dx$.

Solution. We have

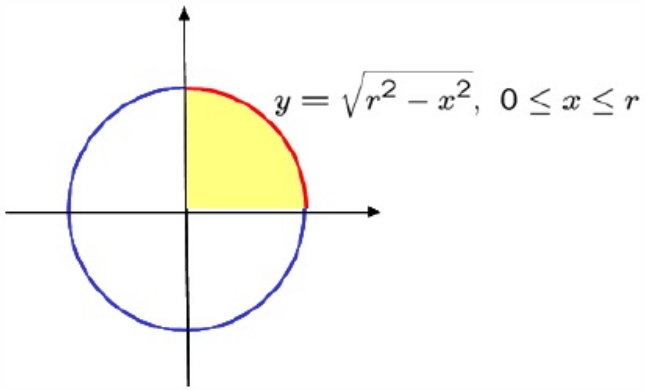
$$\int_0^{\frac{\pi}{2}} \sin(3x)\cos(5x)\,dx = \int_0^{\frac{\pi}{2}} \frac{\sin(-2x)+\sin(8x)}{2}\,dx$$
$$= \frac{1}{2}\left(\int_0^{\frac{\pi}{2}} \sin(8x)\,dx - \int_0^{\frac{\pi}{2}} \sin(2x)\,dx\right)$$
$$= \frac{1}{2}\left(\left[-\frac{\cos(8x)}{8}\right]_0^{\frac{\pi}{2}} - \left[-\frac{\cos(2x)}{2}\right]_0^{\frac{\pi}{2}}\right) = -\frac{1}{2}.$$

(2) Trigonometric Substitution

Example

Show that the area of a circle of radius r is πr^2.

Solution. We place the center of the circle at the origin of the coordinate system and compute the area of the region in the first quadrant.



We let $x = r\sin\theta \implies dx = r\cos\theta\, d\theta$. For $x = 0 \to r$, we have $\theta = 0 \to \dfrac{\pi}{2}$. Hence,

$$\int_0^{\frac{\pi}{2}} \sqrt{r^2 - x^2}\, dx = \int_0^{\frac{\pi}{2}} \sqrt{r^2 - r^2\sin^2\theta}\, (r\cos\theta\, d\theta)$$

$$= \int_0^{\frac{\pi}{2}} \sqrt{r^2(1-\sin^2\theta)}\, (r\cos\theta)\, d\theta = \int_0^{\frac{\pi}{2}} |r| \cdot |\cos\theta| \cdot (r\cos\theta)\, d\theta$$

$$\underbrace{=}_{\text{(because } r > 0 \text{ and } \cos\theta \geq 0)} r^2 \int_0^{\frac{\pi}{2}} \cos^2\theta\, d\theta = \frac{\pi r^2}{4}.$$

So the area of the circle is $4 \cdot \dfrac{\pi r^2}{4} = \pi r^2$.

The following trigonometric substitutions are often used:

- For the expression $\sqrt{a^2 - x^2}$, use the substitution $x = a\sin\theta$, $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.
- For the expression $\sqrt{a^2 + x^2}$, use the substitution $x = a\tan\theta$, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$.
- For the expression $\sqrt{x^2 - a^2}$, use the substitution $x = a\sec\theta$, $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$.

(3) Integration by partial fractions

Now we study how to use the method of partial fractions to help us find the anti-derivative of rational functions. The topics are:

1. How to factorize a polynomial.
2. How to decompose a given rational function into proper partial fractions.
3. How to integrate different kinds of proper partial fractions.

- **Factorizing of polynomials**

A polynomial is said to be irreducible if it cannot be factorized into two or more polynomials (with real coefficients) of smaller degree.

For instance, $p(x) = x^2 + x + 1$ is irreducible because we cannot find two linear polynomials whose product gives $p(x)$, while $q(x) = x^2 - 2$ is reducible because we can factorize

$$x^2 - 2 = \left(x + \sqrt{2}\right)\left(x - \sqrt{2}\right).$$

The following is an important theorem from algebra.

Theorem

Any polynomial can be factorized as a product of linear factors and irreducible quadratic factors.

Remarks.

- A quadratic polynomial $ax^2 + bx + c$ is irreducible if and only if its discriminant $b^2 - 4ac < 0$.
- Every polynomial of degree ≥ 3 is reducible. We can keep factorizing a polynomial into factors of smaller degree until we reach a linear or an irreducible quadratic factor.

- **Rational functions**

A rational function is defined as the ratio of two polynomials. That is, a rational function may be written in the form $f(x) = \dfrac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials. For instance,

$$\frac{2x^2 - x + 4}{x^3 + 4x} \qquad \text{and} \qquad \frac{x^3 + x}{x - 1}$$

are both rational functions.

A rational function is said to be proper if the degree of the numerator is strictly less than the degree of the denominator; otherwise, the rational function is improper.

In the example above the first rational function is proper while the second is improper.

- **Partial fractions decomposition of proper rational functions**

Theorem (Partial Fractions Decomposition)

Any proper rational function $f(x) = \dfrac{p(x)}{q(x)}$ *(i.e.* $\deg(p) < \deg(q)$*) can be expressed as a sum of partial fractions of the form*

$$\frac{A}{(ax+b)^i} \qquad \text{or} \qquad \frac{Ax+B}{(ax^2+bx+c)^j},$$

where $ax + b$ *is a linear factor of* $q(x)$ *and* $ax^2 + bx + c$ *is an irreducible quadratic factor of* $q(x)$.

More information about the theorem above.

- If $ax + b$ is a linear factor of $q(x)$ which is repeated r times in the factorization of $q(x)$, then we have to add the following partial fractions to the partial fractions decomposition of $\frac{p(x)}{q(x)}$:

$$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_r}{(ax+b)^r}.$$

- If an irreducible quadratic factor $ax^2 + bx + c$ of $q(x)$ is repeated s times in the factorization of $q(x)$, then we have to add the following partial fractions to the partial fractions decomposition of $\frac{p(x)}{q(x)}$:

$$\frac{A_1x + B_1}{ax^2+bx+c} + \frac{A_2x + B_2}{(ax^2+bx+c)^2} + \cdots + \frac{A_sx + B_s}{(ax^2+bx+c)^s}.$$

The process to decompose a given proper rational function $f(x) = \dfrac{p(x)}{q(x)}$ into partial fractions is always the same.

The steps are as follow.

1. Factorize the denominator $q(x)$ completely into linear factors and irreducible quadratic factors.
2. For each linear factor $ax + b$ of $q(x)$, see how many times $ax + b$ is repeated. Add the corresponding terms

$$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_r}{(ax+b)^r}.$$

3. For each irreducible quadratic factor $ax^2 + bx + c$ of $q(x)$, see how many times $ax^2 + bx + c$ is repeated. Add the corresponding terms

$$\frac{A_1 x + B_1}{ax^2 + bx + c} + \frac{A_2 x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_s x + B_s}{(ax^2 + bx + c)^s}.$$

   Use different symbols for the constants.

4. Once you write down all the different terms in the partial fractions decomposition, combine all the terms into a single fraction. Compare coefficients with $\frac{p(x)}{q(x)}$. Get a system of simultaneous equations.

5. Solve the system of simultaneous equations to obtain the values of the constants $A_1, B_1, \ldots$.

Example

Find the partial fractions decomposition of $f(x) = \dfrac{2x^2 - x + 4}{x^3 + 4x}$.

Solution. We follow the steps above.

1) Factorize the denominator $q(x) = x^3 + 4x$ completely into linear factors and irreducible quadratic factors: $x^3 + 4x = x(x^2 + 4)$.

2) There is only one linear factor, x, which is repeated once. So we add the term $\dfrac{A}{x}$.

3) There is only one quadratic factor, $x^2 + 4$, which is repeated once. So we add the term $\dfrac{Bx + C}{x^2 + 4}$.

4) The terms are

$$\frac{2x^2 - x + 4}{x^3 + 4x} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}.$$

So this means that

$$\frac{2x^2 - x + 4}{x^3 + 4x} = \frac{(A + B)x^2 + Cx + 4A}{x(x^2 + 4)}.$$

Equating the coefficients of x^2, x and the constant term in the numerator on both sides, we get the following system of simultaneous equations:

$$A + B = 2, \qquad C = -1, \qquad 4A = 4.$$

5) Solving this system of simultaneous equations, we see that the solution is

$$A = 1, \qquad B = 1, \qquad C = -1.$$

Therefore, the partial fractions decomposition of $f(x)$ is

$$\frac{2x^2 - x + 4}{x^3 + 4x} = \frac{1}{x} + \frac{x - 1}{x^2 + 4}.$$

The purpose of doing partial fractions is to find the integral of a given rational function.

- Often it is difficult to integrate a rational function $\frac{p(x)}{q(x)}$ directly, but if the given rational function is proper, we can express it in partial fractions, and then integrate the partial fractions term by term (which is normally an easier task than integrating $\frac{p(x)}{q(x)}$ directly).

For instance, it is not easy to see how to integrate $\frac{2x^2 - x + 4}{x^3 + 4x}$ directly, but it is easy to integrate $\frac{1}{x} + \frac{x - 1}{x^2 + 4}$.

- So we now study how to integrate a fraction with a linear denominator $ax + b$ and an irreducible quadratic denominator $ax^2 + bx + c$.

- **Integrating rational functions**
  - Proper rational functions

Note that to integrate proper rational functions, the following are often used:

$$\int \frac{1}{x}\, dx = \ln|x| + C,$$
$$\int \frac{1}{x^2 + a^2}\, dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C,$$

where C is an arbitrary constant.

You might want to memorize these two formulas.

Example

Evaluate $\displaystyle\int_1^2 \frac{1}{x^5+2x^3+x}\,dx$.

Solution. First, factorize $x^5+2x^3+x = x(x^2+1)^2$, and hence decompose

$$\frac{1}{x^5+2x^3+x} = \frac{1}{x} - \frac{x}{x^2+1} - \frac{x}{(x^2+1)^2}.$$

Integrating it, we get

$$\begin{aligned}
\int_1^2 \frac{1}{x^5+2x^3+x}\,dx &= \int_1^2 \frac{1}{x}\,dx - \int_1^2 \frac{x}{x^2+1}\,dx - \int_1^2 \frac{x}{(x^2+1)^2}\,dx \\
&= \Big[\ln|x|\Big]_1^2 - \frac{1}{2}\Big[\ln|u|\Big]_2^5 - \frac{1}{2}\Big[-\frac{1}{u}\Big]_2^5 \quad (\text{here } u = x^2+1) \\
&= \ln 2 - \frac{1}{2}\Big(\ln 5 - \ln 2\Big) - \frac{3}{20} = \frac{3}{2}\ln 2 - \frac{1}{2}\ln 5 - \frac{3}{20}.
\end{aligned}$$

- Improper rational functions

If we are given an improper rational function $\frac{p(x)}{q(x)}$ $(\deg(p) \geq \deg(q))$, we must

- First perform a long division

$$\frac{p(x)}{q(x)} = r(x) + \frac{s(x)}{q(x)},$$

where $r(x)$ is a polynomial and $\frac{s(x)}{q(x)}$ is a proper rational function.

- Then proceed to decompose $\frac{s(x)}{q(x)}$ into partial fractions and to integrate the whole expression $r(x) + \frac{s(x)}{q(x)}$.

Outline

1 Integration Techniques

2 Improper Integrals

Recall that a definite integral $\int_a^b f(x)\,dx$ is introduced under two assumptions:

- First, the length of the interval $[a, b]$ is required to be bounded.
- Second, the function $f(x)$ is required to be bounded on the interval $[a, b]$.

In simpler terms, we may say that a definite integral requires both the domain and the range of the integrand $f(x)$ to be bounded.

Therefore, for none of the following can the definite integral be defined:

$$\int_1^{\infty} \frac{1}{x^2}\,dx, \quad \int_{-\infty}^{0} xe^x\,dx, \quad \int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx,$$
$$\int_0^2 \frac{1}{\sqrt{2-x}}\,dx, \quad \int_0^1 \ln x\,dx, \quad \int_0^3 \frac{1}{(x-1)^{\frac{2}{3}}}\,dx.$$

Each case above is called an improper integral.

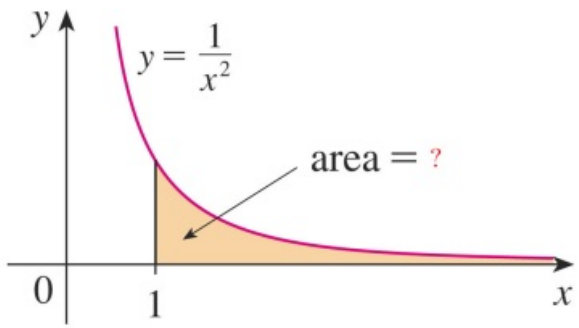
In this section we focus on the study of improper integrals. One of the most important applications of improper integrals lies in probability and statistics.

• Illustrative examples

Let's begin with the discussion of the following improper integral whose upper limit is ∞:

$$\int_1^{\infty} \frac{1}{x^2}\, dx.$$

Since the graph of $\dfrac{1}{x^2}$ is positive, we expect this improper integral to have the same value as the area of the region S under the curve $y = \dfrac{1}{x^2}$ and to the right of the line $x = 1$.



As we can see from the figure, this region S is infinite in some sense, since it is unbounded on the right hand side.

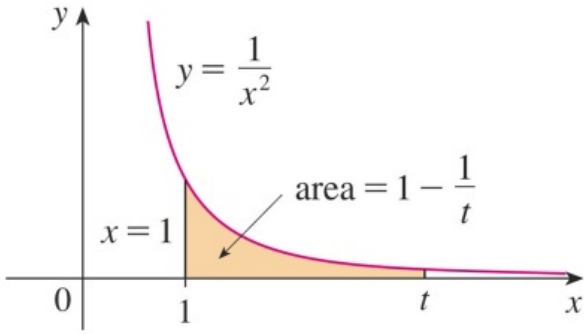
This might lead us to guess that the value of the area of S should be ∞ as well.

Unfortunately this is not true.

To see why, we need to ask ourselves:

- *How can we measure the area of a region that is infinite in the sense of being unbounded?*
- *Can such a region be unbounded and yet have finite area?*

The answer to this lies, yet again, in the process of taking limits. We first fix a number $t > 1$ and measure the area under the curve between $x = 1$ and $x = t$.



Since this region is now bounded (the interval $[1, t]$), we can compute this area by taking the definite integral:

$$A(t) = \int_1^t \frac{1}{x^2}\, dx = \left[-\frac{1}{x}\right]_1^t = 1 - \frac{1}{t}.$$

Note that this area depends on the t we choose and so $A(t)$ should be a function of t.

Now what conclusion can we arrive at?

As t increases to infinity, the area of the bounded region should approach the area of the unbounded region. So we can define the area of the unbounded region as follows:

$$\int_1^{\infty} \frac{1}{x^2}\, dx = \lim_{t\to\infty} A(t) = \lim_{t\to\infty} \left(1 - \frac{1}{t}\right) = 1.$$

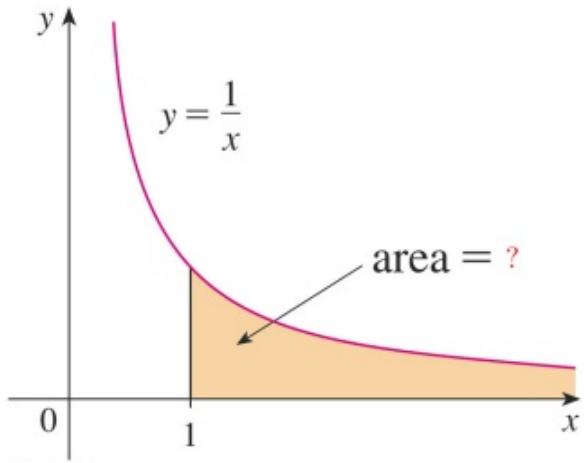
This shows that the area of an unbounded region can be a finite value!

The example above gives us a method to define improper integrals: we should take limits whenever appropriate.

Now let's apply the method to try and evaluate another improper integral:

$$\int_1^{\infty} \frac{1}{x}\, dx.$$

This improper integral is expected to have a value equal to the area of the shaded region in the figure



We can compute the integral in the same way. That is,

$$\int_1^{\infty} \frac{1}{x}\, dx = \lim_{t\to\infty} \int_1^{t} \frac{1}{x}\, dx = \lim_{t\to\infty} \Big[\ln x\Big]_1^{t} = \lim_{t\to\infty} \ln t = \infty.$$

It's interesting!

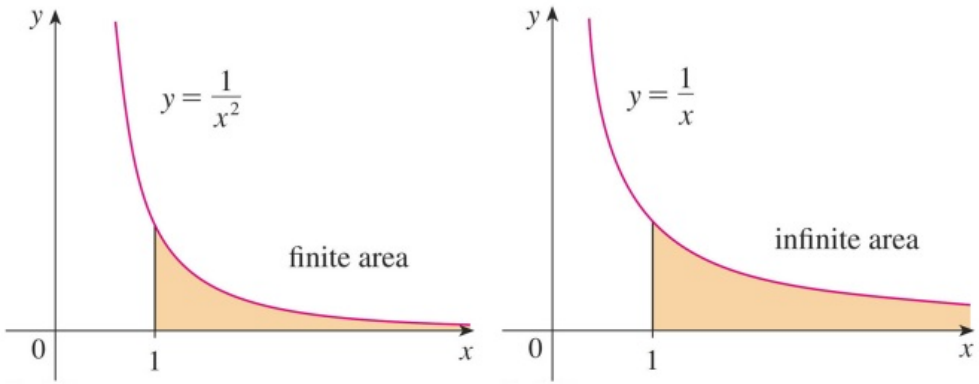
This limit is ∞ and does not give a finite value.

The interpretation of this answer is that the unbounded region above has infinite area.

Let's compare geometrically

$$\int_1^\infty \frac{1}{x^2}\,dx = 1 \qquad \text{versus} \qquad \int_1^\infty \frac{1}{x}\,dx = \infty.$$

Although the curves $y = \dfrac{1}{x^2}$ and $y = \dfrac{1}{x}$ look very similar, the area under the first curve (the second curve) has finite value (infinite value).



Note that in this figure, the left graph is lower than the right graph.

So an unbounded region can have finite or infinite area! This means that an improper integral of the form $\int_a^{\infty} f(x)\,dx$ or $\int_{-\infty}^{a} f(x)\,dx$ can have finite or infinite limit.

- If this limit exists and is finite, then we say that the improper integral converges.
- If this limit does not exist, or is infinite, then we say that the improper integral diverges.

(1) Improper integrals of type I

Attention! *You should be able to recognize the improper integrals of this type (Type I) and know how to evaluate them.*

There are three types of an improper integral of type I:

(a) If f is continuous on the interval $[a, \infty)$, then

$$\int_a^{\infty} f(x)\, dx = \lim_{t \to \infty} \int_a^t f(x)\, dx,$$

provided that the limit on the right exists and is finite.

(b) If f is continuous on the interval $(-\infty, b]$, then

$$\int_{-\infty}^{b} f(x)\, dx = \lim_{t \to -\infty} \int_t^b f(x)\, dx,$$

provided that the limit on the right exists and is finite.

(c) If f is continuous on $(-\infty, \infty)$ then

$$\int_{-\infty}^{\infty} f(x)\, dx = \int_{-\infty}^{c} f(x)\, dx + \int_{c}^{\infty} f(x)\, dx,$$

provided that both limits $\int_{-\infty}^{c} f(x)\, dx$ and $\int_{c}^{\infty} f(x)\, dx$ exist and are both finite. Here c can be taken to be any real number.

Some remarks.

(1) In each case above we need the corresponding limit to exist and to be finite, i.e. the improper integral is convergent. If the corresponding limit does not exist (e.g. oscillating) or is infinite, then the improper integral is divergent.

(2) In case (c) above we need both limits to exist and to be finite. Thus we may write

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_{c}^{\infty} f(x)\,dx$$
$$= \left(\lim_{t\to-\infty} \int_{t}^{c} f(x)\,dx\right) + \left(\lim_{t\to\infty} \int_{c}^{t} f(x)\,dx\right)$$

If one (or both) of the limits is oscillating or infinite, then the whole improper integral $\int_{-\infty}^{\infty} f(x)\,dx$ is divergent.

(3) Thus in case (c) we need to evaluate two different limits

$$\lim_{t\to-\infty}\int_t^c f(x)\,dx \quad \text{and} \quad \lim_{t\to\infty}\int_c^t f(x)\,dx.$$

This is not the same as evaluating

$$\lim_{t\to\infty}\int_{-t}^{t} f(x)\,dx;$$

this is not what we should evaluate. In general,

$$\int_{-\infty}^{\infty} f(x)\,dx \neq \lim_{t\to\infty}\int_{-t}^{t} f(x)\,dx.$$

Example

Show that $\int_{-\infty}^{\infty} x\,dx \neq \lim_{t\to\infty}\int_{-t}^{t} x\,dx$.

Solution. We first work out the right-hand side:

$$\lim_{t\to\infty}\int_{-t}^{t} x\,dx = \lim_{t\to\infty}\left[\frac{x^2}{2}\right]_{-t}^{t} = \lim_{t\to\infty}\left[\frac{t^2}{2} - \frac{(-t)^2}{2}\right]$$
$$= \lim_{t\to\infty} 0 = 0.$$

To evaluate $\int_{-\infty}^{\infty} x\,dx$, we need to follow the definition. We can take c to be any real number, so let's take $c = 0$.

We have

$$\int_{-\infty}^{\infty} x\,dx = \int_{-\infty}^{0} x\,dx + \int_{0}^{\infty} x\,dx$$
$$= \left(\lim_{t\to-\infty}\int_{t}^{0} x\,dx\right) + \left(\lim_{t\to\infty}\int_{0}^{t} x\,dx\right)$$
$$= \lim_{t\to-\infty}\left[\frac{x^2}{2}\right]_{t}^{0} + \lim_{t\to\infty}\left[\frac{x^2}{2}\right]_{0}^{t} = \lim_{t\to-\infty}\left(-\frac{t^2}{2}\right) + \lim_{t\to\infty}\frac{t^2}{2}.$$

It is clear that both limits are divergent

$$\lim_{t\to-\infty}\left(-\frac{t^2}{2}\right) = -\infty \quad \text{and} \quad \lim_{t\to\infty}\frac{t^2}{2} = \infty.$$

So by definition of the improper integral, we conclude that $\int_{-\infty}^{\infty} x\,dx$ is divergent.

- Note that we cannot do

$$\lim_{t\to-\infty}\left(-\frac{t^2}{2}\right)+\lim_{t\to\infty}\frac{t^2}{2}=(-\infty)+\infty=0$$

to conclude that the improper integral converges to 0!

- You must check separately if each limit converges. In fact, you should recognize $\infty-\infty$ to be an indeterminate form, because it can take on any possible value. So we cannot say that $(-\infty)+\infty=0$.

- It is also incorrect to say that

$$\lim_{t\to-\infty}\left(-\frac{t^2}{2}\right)+\lim_{t\to\infty}\frac{t^2}{2}=\lim_{t\to\infty}\left(-\frac{t^2}{2}\right)+\lim_{t\to\infty}\frac{t^2}{2}$$

$$\lim_{t\to\infty}\left(-\frac{t^2}{2}+\frac{t^2}{2}\right)=\lim_{t\to\infty}0=0.$$